

Vishakha Singh

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Professional Summary

AI/ML Engineer with 9+ years of software engineering experience, building Generative AI applications, AI agents, Retrieval-Augmented Generation (RAG) systems, and production-ready ML pipelines using Python. Experienced with Azure OpenAI, LangGraph, LangChain, FastAPI, and modern LLM application development while leveraging a strong background in enterprise-scale software engineering.

Core Skills

AI / ML & Generative AI

- LLMs, RAG (Retrieval-Augmented Generation), NLP, LangChain, LangGraph
- Transformers, Text Embeddings, Prompt Engineering
- Model Evaluation & Optimization

Frameworks & Libraries

- PyTorch, TensorFlow
- Hugging Face Transformers
- Scikit-learn, Pandas, NumPy

MLOps & Deployment

- FastAPI for AI inference APIs
- Model versioning & monitoring

Cloud & AI Platforms

- Azure OpenAI (enterprise authentication, deployment)

Programming

- Python
- SQL (Oracle, DB2, MySQL)

Tools

- Git, VS Code, PyCharm
- Postman, ServiceNow

- Data processing: Shell scripting, Linux
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AI/ML Project Experience

◆ Intelligent AI Support Agent using LangGraph

- Built a stateful AI support agent using LangGraph and StateGraph for multi-step LLM workflows.
- Designed graph-based workflows with conditional routing and persistent conversation state.
- Implemented checkpointing and memory for workflow recovery and contextual conversations.
- Developed modular retrieval, reasoning, and response generation components using Azure OpenAI.
- Designed a reusable architecture supporting future tool calling and human-in-the-loop workflows.

Tech Stack: Python, LangGraph, LangChain, Azure OpenAI, StateGraph, FastAPI, Prompt Engineering, Memory, Checkpointing

◆ Titanic Survival Prediction (Machine Learning & Model Deployment)

- Built an end-to-end machine learning pipeline to predict passenger survival using structured data.
- Implemented and compared Logistic Regression, Random Forest, XGBoost, and PyTorch Neural Networks.
- Performed feature engineering, hyperparameter tuning, and model evaluation using Accuracy, ROC-AUC, Precision/Recall, and Confusion Matrix.
- Developed a FastAPI inference service for real-time predictions.

Tech Stack: Python, Scikit-learn, XGBoost, PyTorch, FastAPI, Pandas

◆ Spam Detection System (ML + LLM-based)

- Developed a spam detection system using traditional machine learning and LLM-based classification approaches.
- Implemented a supervised ML classifier using text preprocessing and feature extraction techniques.
- Integrated Azure OpenAI/LLM with prompt engineering for zero-shot and few-shot spam classification.
- Evaluated traditional ML and LLM approaches across accuracy, inference quality, and real-world usability.

- Designed the solution as a modular pipeline for real-time inference.

Tech Stack: Python, Scikit-learn, Pandas, Azure OpenAI, Prompt Engineering, FastAPI

Professional Experience

Senior Software Engineer – Optum (UHG)

 Jul 2020 – Present

- Developed Python-based automation solutions for large-scale data validation, log analysis, and workflow optimization, improving operational efficiency.
 - Built scalable data processing and reconciliation pipelines for enterprise datasets using Python and SQL.
 - Applied AI/ML concepts, including anomaly detection and pattern analysis, to enhance system monitoring and automate issue identification.
 - Designed reusable, modular automation components that reduced manual effort and supported scalable data-driven workflows.
 - Collaborated with cross-functional teams to modernize legacy systems by integrating Python-based automation and AI-ready solutions.
 - Led incident analysis and developed intelligent automation tools to improve operational reliability and accelerate root cause analysis
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Senior Systems Engineer – Infosys (UBS Bank)

 Jun 2017 – Jun 2020

- Developed and maintained large-scale enterprise systems handling financial data
 - Performed system design, testing, and production support
 - Led impact analysis and collaborated across global teams
 - Strong foundation in high-reliability systems, data engineering, and architecture design
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Education

B.Tech – BPUT (2017) – 9.0 CGPA

Achievements

- AI Dojo Program – Completed with Excellent Grade
- Multiple Spot Awards (Infosys + Optum)
- Recognized for automation and incident resolution excellence